Lab Report 02

Course title: Computer Network Laboratory
Course code: CSE-3202
Session: July - December 2026

Date of Submission: *19 August,2026*



Submitted to-

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Title: Manual Router Configuration and Static Routing Using Cisco Packet Tracer

Objective:

- To configure router interfaces manually in Cisco Packet Tracer.
- To assign IP addresses and default gateways to PCs.
- To connect two different LANs using a router.
- To configure static routes between two routers.

Description:

Part 1: Manual Router Configuration

In this experiment, one 2911 router, two 2960 switches and four PCs were used. Two PCs were connected to each switch, creating two separate LANs.

For the first LAN, the PCs were assigned the following IPv4 addresses:

- PC1: 10.0.0.1
- PC2: 10.0.0.2
- Default Gateway: 10.0.0.3

The GigabitEthernet 0/0 interface of the router was configured with the IP address 10.0.0.3. The port status was also turned on.

For the second LAN, the PCs were assigned:

- PC3: 192.168.0.1
- PC4: 192.168.0.2
- Default Gateway: 192.168.0.3

The GigabitEthernet 0/1 interface of the router was configured with the IP address 192.168.0.3, and the port was turned on.

After completing the configuration, connectivity was tested using the ping command. The ping between PCs in the same LAN was successful. The ping between PCs from the two different LANs was also successful. This confirmed that the router was correctly configured and was allowing communication between the two networks.

Part 2: Static Routing

For the static routing experiment, two 2911 routers were used. Router 0 was connected to one switch, while Router 1 was connected to two switches. Each LAN contained two PCs.

First the PCs were configured with IPv4 addresses, subnet masks and default gateways according to their respective LANs. The router interfaces were also configured with the required IP addresses and enabled.

The two routers were connected through their GigabitEthernet interfaces. GigabitEthernet 0/1 of Router 0 was assigned the IP address 192.168.2.1, while GigabitEthernet 0/0 of Router 1 was assigned the IP address 192.168.2.2. These two interfaces belong to the same network and provide communication between Router 0 and Router 1.

After completing the basic router configuration, static routes were added to the routers. This was done from Config - Routing - Static. For each route, the destination Network, Mask and Next Hop were specified. The next-hop address was the IP address of the connected interface of the other router.

The static routes allowed each router to forward packets to networks that were not directly connected to it. Finally, connectivity was tested using the ping command. Successful ping results confirmed that the routers could communicate with each other and that the static routing configuration was working correctly.

Connection:

Part 1: Manual Router Configuration

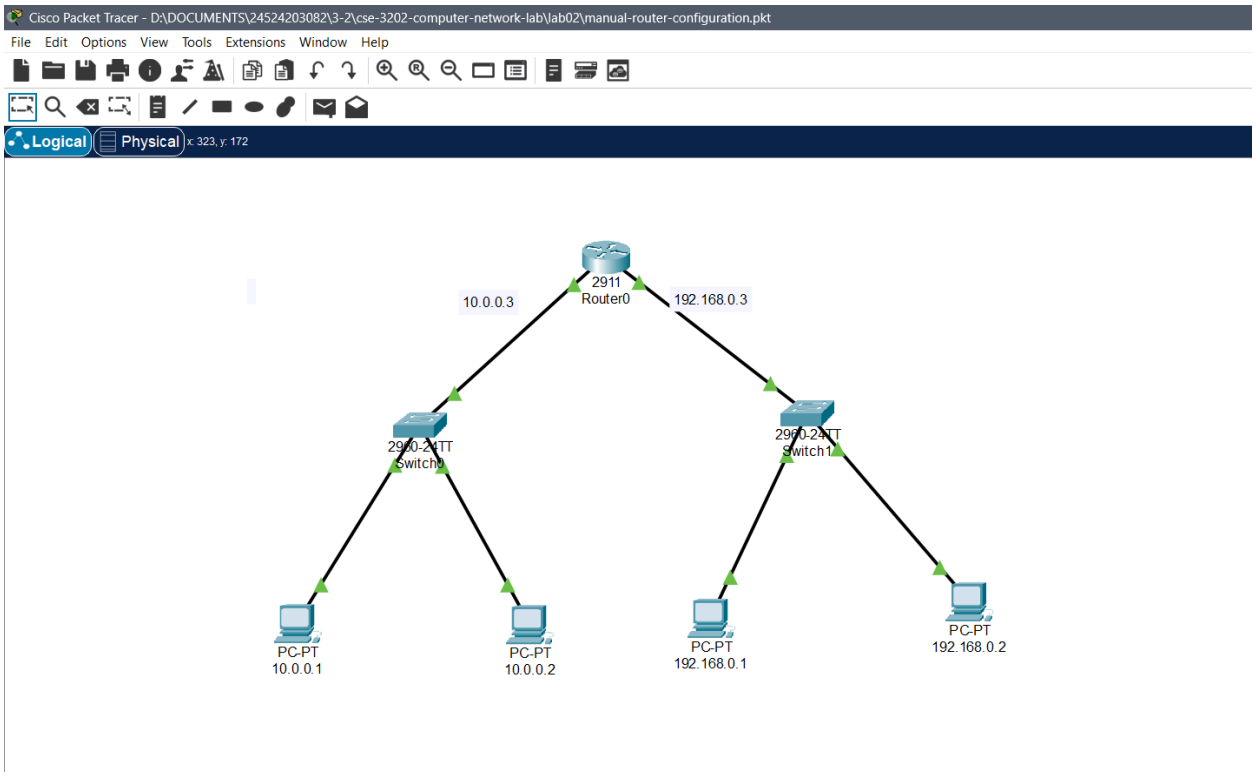


Figure 1: Network Topology for Manual Router Configuration

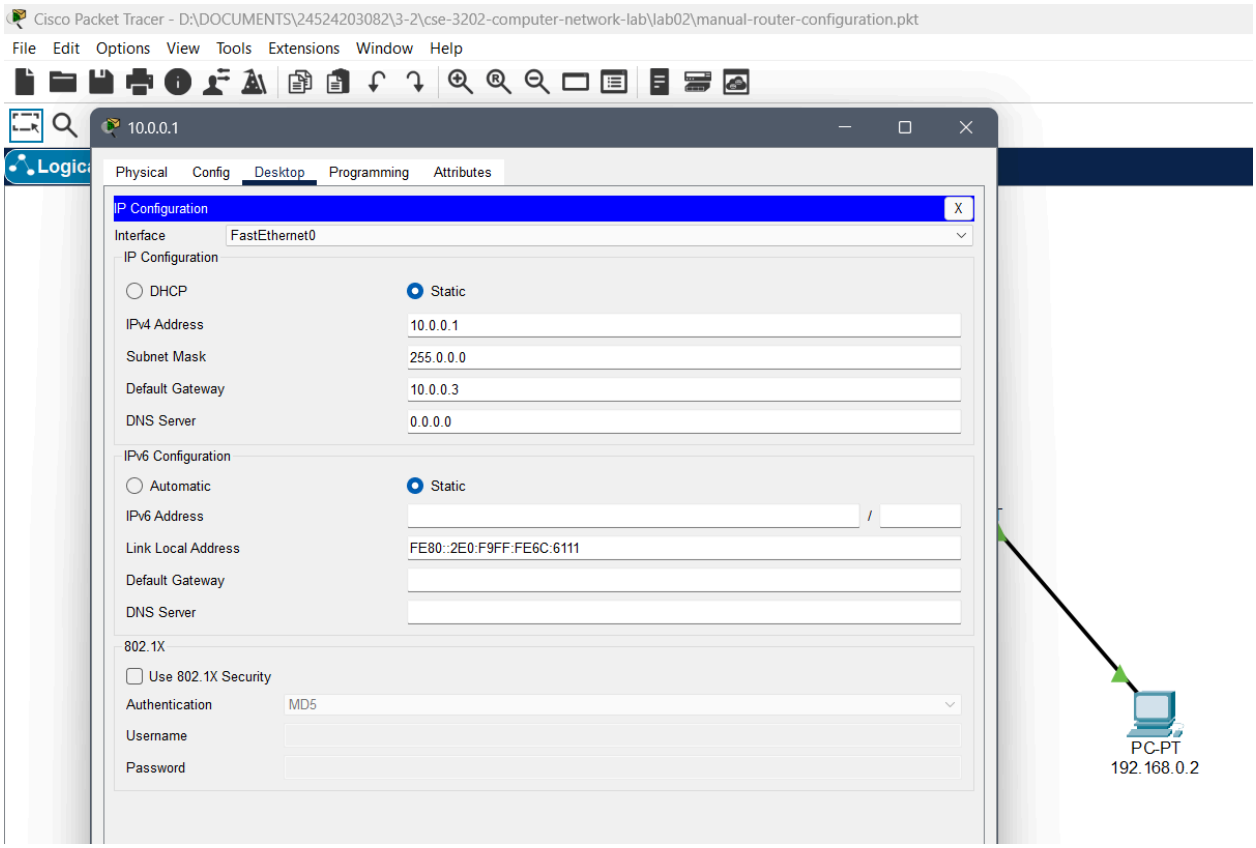


Figure 2: IP Configuration of PC in the First LAN

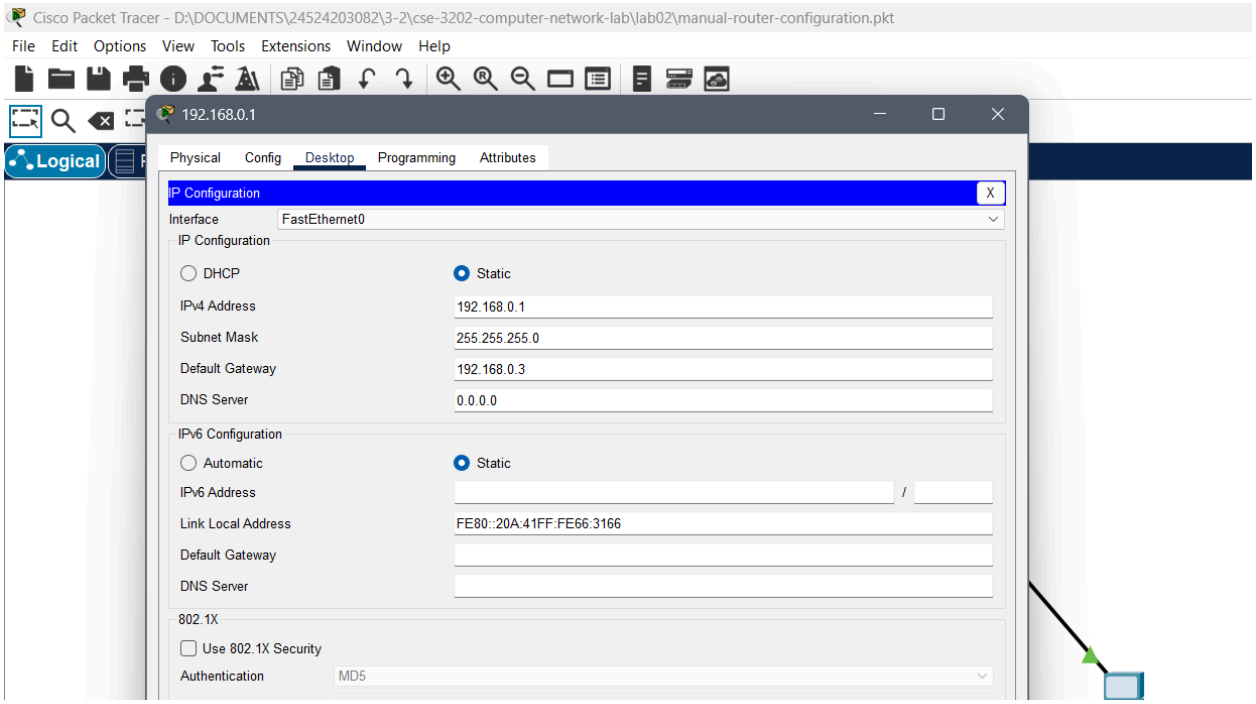


Figure 3: IP Configuration of PC in the Second LAN

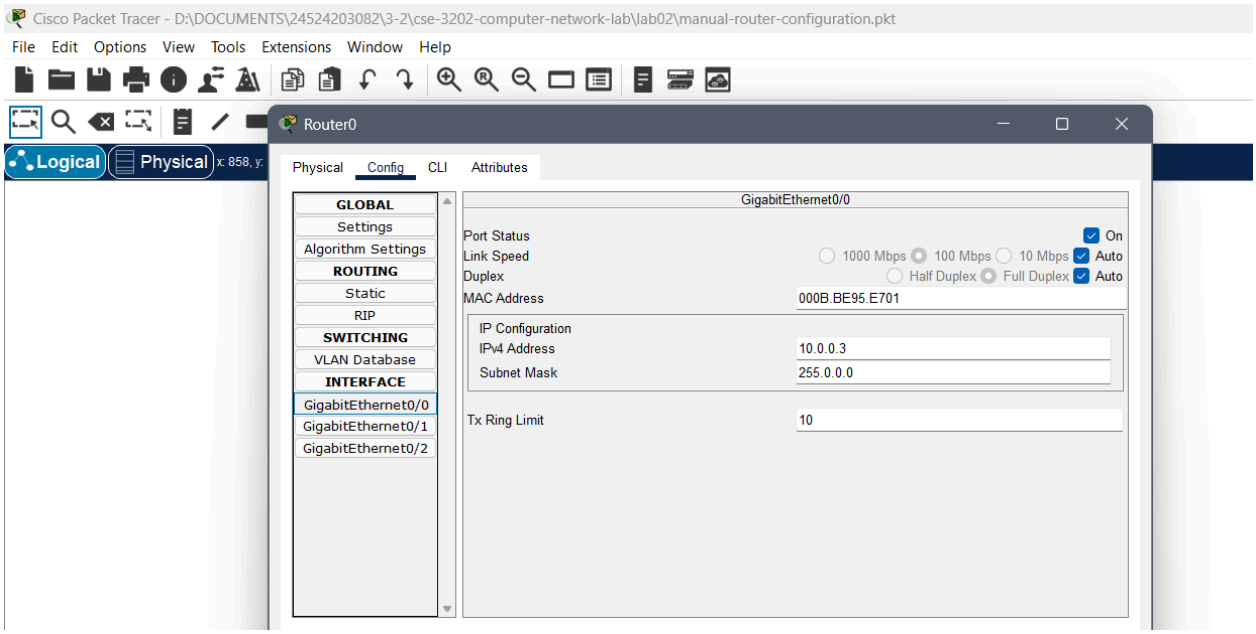


Figure 4: GigabitEthernet 0/0 Router Configuration

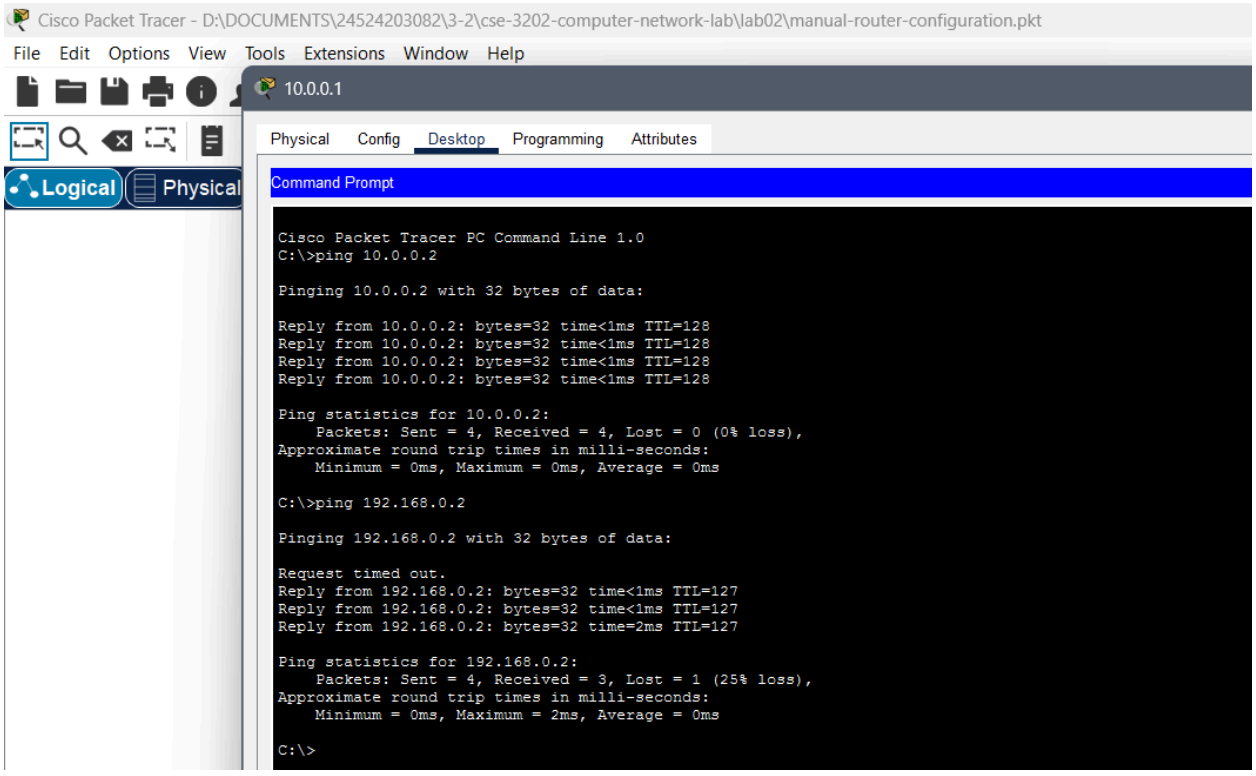


Figure 5: Command Prompt Showing Successful Ping in the Same LAN and between Different LANs

Part 2: Static Routing

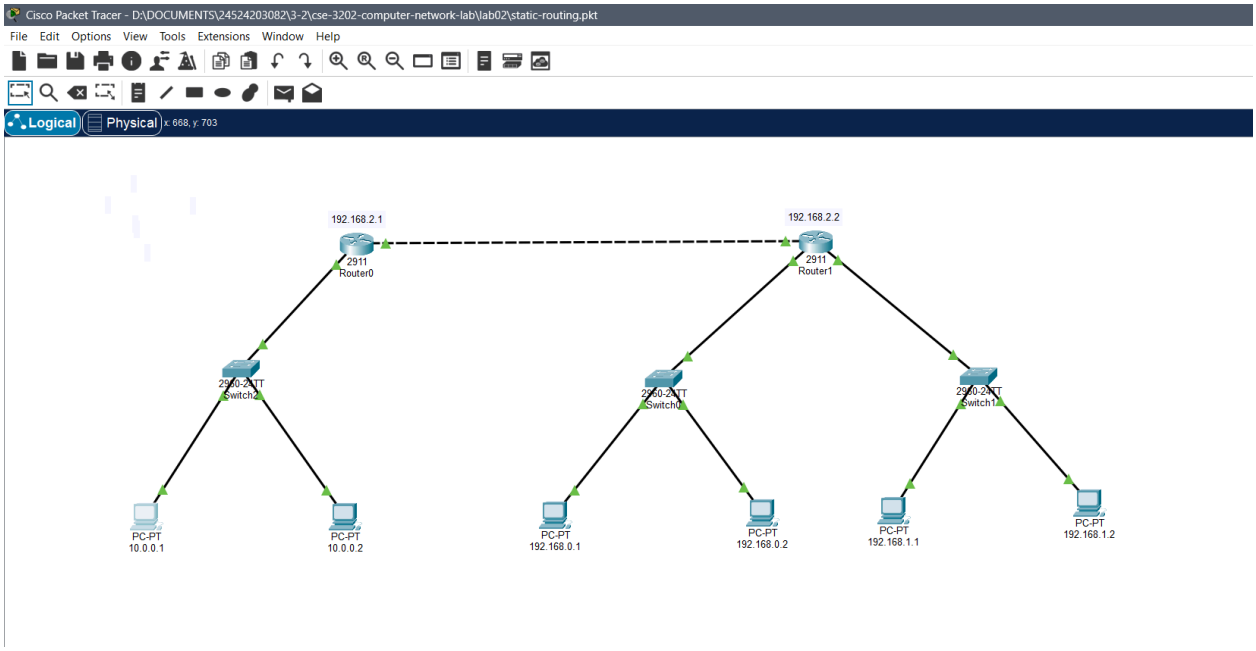


Figure 6: Network Topology for Static Routing

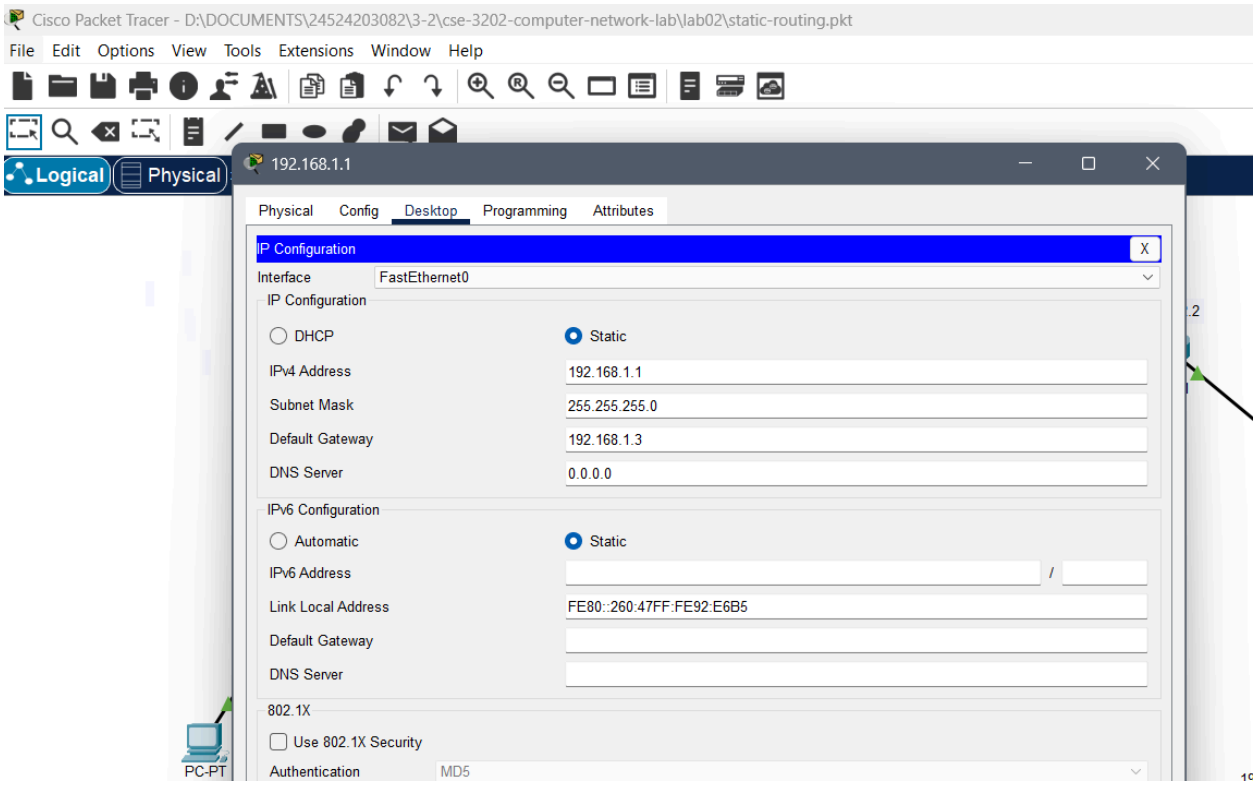


Figure 7: IP Configuration of PC in the Static Routing Network

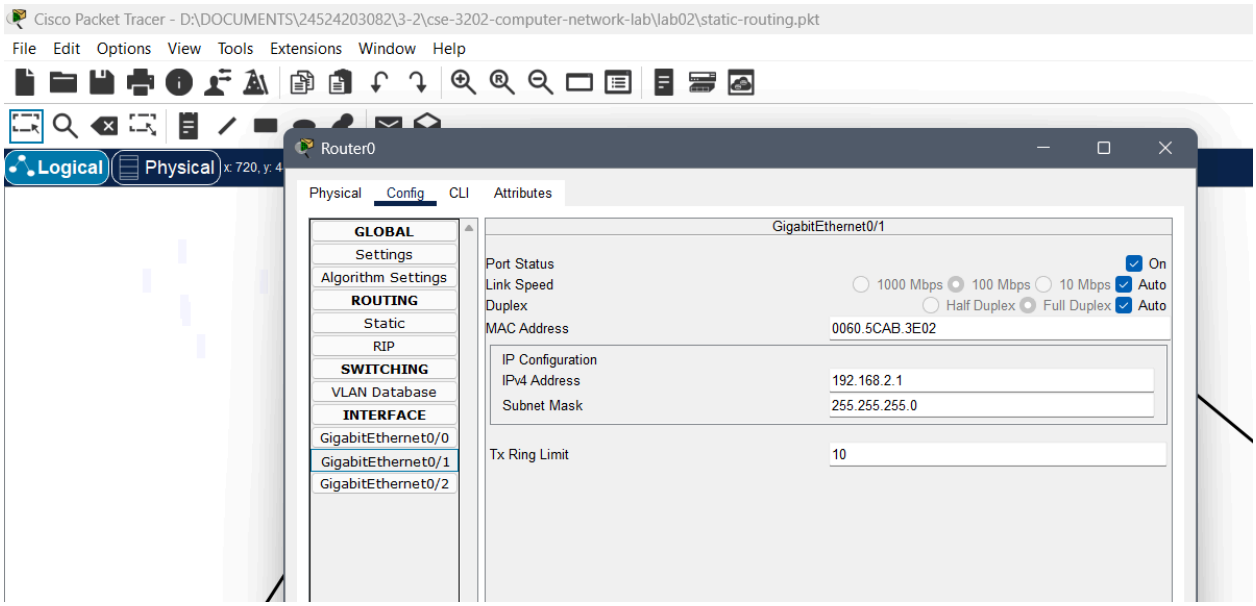


Figure 8: Router Interface Configuration

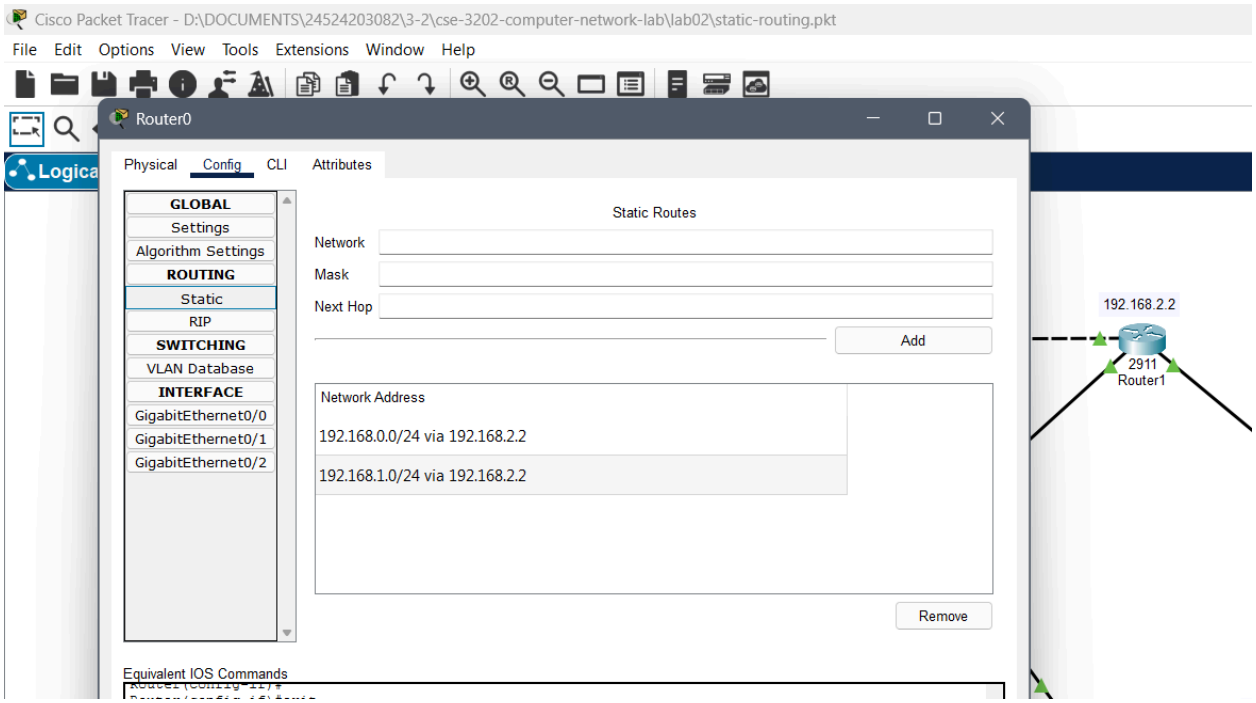


Figure 9: Static Route Configuration

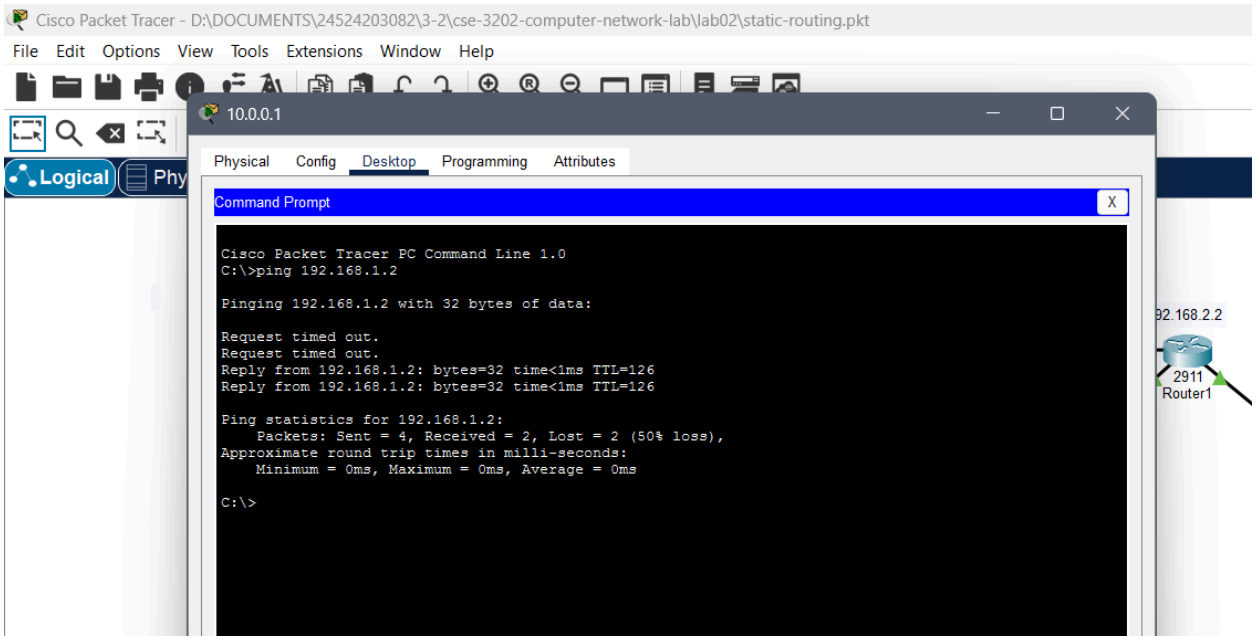


Figure 10: Command Prompt Showing Successful Ping